PROBLEM2:

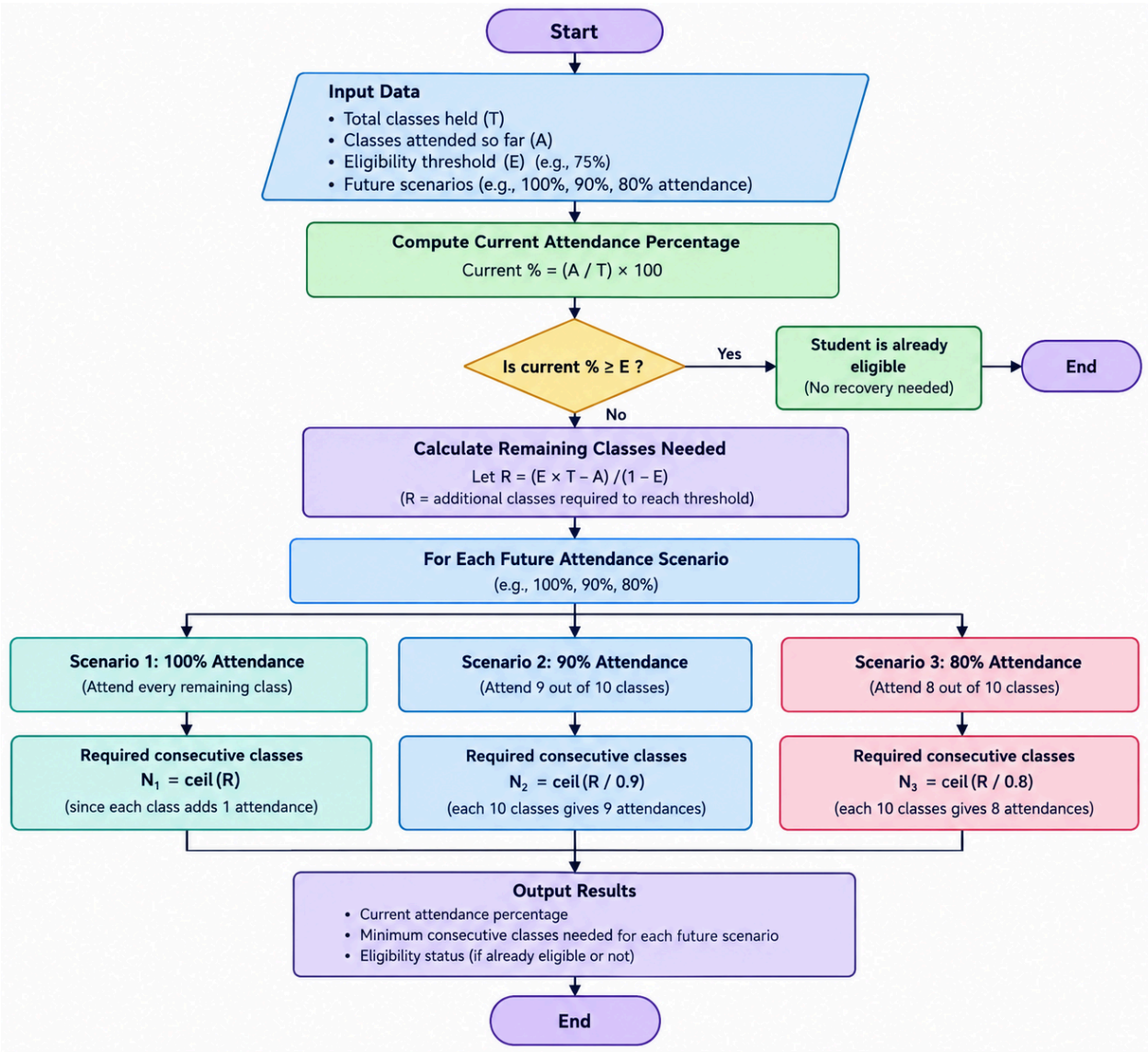
PROBLEM STATEMENT:

Attendance Recovery Computation: Calculate current attendance percentages and determine the minimum consecutive classes required to reach an eligibility threshold under different future attendance scenarios.

ALGORITHM:

1. Start.
2. Read the total number of classes conducted (`total`).
3. Read the number of classes attended (`attended`).
4. Read the required eligibility percentage (`threshold`).
5. Calculate the current attendance percentage:
   \[\text{percentage}=\frac{\text{attended}}{\text{total}}\times100 \]
6. If the current percentage is greater than or equal to the threshold, display “Eligible”.
7. Otherwise, initialize the number of future classes attended (`future`) to 0.
8. Calculate the new attendance percentage after attending future classes:
   \[\text{new percentage}=\frac{\text{attended}+\text{future}}{\text{total}+\text{future}}\times100 \]
9. Increase `future` by 1 until the new percentage reaches the required threshold.
10. Display the minimum number of consecutive classes required.
11. Stop.

FLOWCHART:



PSEUDOCODE:

BEGIN

INPUT total
INPUT attended
INPUT threshold

```
    IF total <= 0 OR attended < 0 OR attended > total THEN
        PRINT "Invalid attendance details"
    ELSE
        percentage = (attended / total) * 100

        PRINT "Current Attendance =", percentage, "%"

        IF percentage >= threshold THEN
            PRINT "Eligible"
            PRINT "Additional classes required = 0"
        ELSE
            future = 0

            WHILE ((attended + future) / (total + future)) * 100 < threshold DO
                future = future + 1
            END WHILE

            PRINT "Not currently eligible"
            PRINT "Minimum consecutive classes to attend =", future
        END IF
    END IF

END
```

PROGRAM:

```
import math

def calculate_attendance_recovery():
    print("--- Attendance Recovery Computation ---")

    # Input total classes held, attended, and target percentage
    total_classes = int(input("Enter total classes held so far: "))
    attended_classes = int(input("Enter total classes attended so far: "))
    target_percentage = float(input("Enter target eligibility percentage (e.g., 75): "))

    # Handle edge case for zero classes
    if total_classes == 0:
        print("Error: Total classes cannot be zero.")
        return

    # Calculate current attendance percentage
    current_percentage = (attended_classes / total_classes) * 100
    print(f"\nCurrent Attendance Percentage: {current_percentage:.2f}%")
```

```
    # Check if already eligible
    if current_percentage >= target_percentage:
        print("Status: You are already eligible! Minimum consecutive classes required = 0")
        return

    # Validation for impossible targets (target >= 100% when current < 100%)
    if target_percentage >= 100:
        print("Status: Target of 100% or more is impossible to reach if you have missed any
classes.")
        return

    # Formula derivation:
    # Target = (Attended + X) / (Total + X) * 100
    # Target * (Total + X) = 100 * Attended + 100X
    # Target * Total + Target * X = 100 * Attended + 100X
    # Target * Total - 100 * Attended = (100 - Target) * X
    # X = (Target * Total - 100 * Attended) / (100 - Target)

    numerator = (target_percentage * total_classes) - (100 * attended_classes)
    denominator = 100 - target_percentage

    # Use math.ceil because we need whole numbers of classes
    required_classes = math.ceil(numerator / denominator)

    print(f"Status: Below eligibility threshold.")
    print(f"Minimum consecutive classes required to attend: {required_classes}")

if __name__ == "__main__":
    calculate_attendance_recovery()
```

OUTPUT:

```
--- Attendance Recovery Computation ---
Enter total classes held so far: 40
Enter total classes attended so far: 32
Enter target eligibility percentage (e.g., 75): 75

Current Attendance Percentage: 80.00%
Status: You are already eligible! Minimum consecutive classes required = 0
```